# Unit 39: Trigonometric Substitution — Full Solutions

Warm-up

1. $\displaystyle\int \frac{dx}{\sqrt{9-x^2}}$ — this matches $\sqrt{a^2-x^2}$ (with $a=3$), so use $x=3\sin\theta$.

2. $\displaystyle\int \frac{dx}{\sqrt{x^2+16}}$ — this matches $\sqrt{a^2+x^2}$ (with $a=4$), so use $x=4\tan\theta$.

3. $\displaystyle\int \frac{dx}{\sqrt{x^2-25}}$ — this matches $\sqrt{x^2-a^2}$ (with $a=5$), so use $x=5\sec\theta$.

4. $\displaystyle\int \frac{dx}{\sqrt{4-x^2}}$, $x=2\sin\theta$

$dx=2\cos\theta\,d\theta$, $\sqrt{4-x^2}=2\cos\theta$.

$$\int \frac{2\cos\theta\,d\theta}{2\cos\theta}=\int d\theta=\theta+C=\arcsin\left(\frac{x}{2}\right)+C$$

(This matches the direct arcsin formula from Unit 36 — good confirmation that trig substitution gives the same answer.)

5. $\displaystyle\int \frac{dx}{\sqrt{9-x^2}}$, $x=3\sin\theta$

$dx=3\cos\theta\,d\theta$, $\sqrt{9-x^2}=3\cos\theta$.

$$\int d\theta=\theta+C=\arcsin\left(\frac{x}{3}\right)+C$$

6. $\displaystyle\int \sqrt{1-x^2}\,dx$, $x=\sin\theta$

$dx=\cos\theta\,d\theta$, $\sqrt{1-x^2}=\cos\theta$.

$$\int \cos\theta\cdot\cos\theta\,d\theta=\int \cos^2\theta\,d\theta=\frac{\theta}{2}+\frac{\sin(2\theta)}{4}+C=\frac{\theta}{2}+\frac{\sin\theta\cos\theta}{2}+C$$

Convert: $\theta=\arcsin x$, $\sin\theta=x$, $\cos\theta=\sqrt{1-x^2}$.

$$=\frac{1}{2}\arcsin x+\frac{x\sqrt{1-x^2}}{2}+C$$

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Standard

7. $\displaystyle\int \frac{x^2}{\sqrt{4-x^2}}\,dx$, $x=2\sin\theta$

$x^2=4\sin^2\theta$, $\sqrt{4-x^2}=2\cos\theta$, $dx=2\cos\theta\,d\theta$.

$$\int \frac{4\sin^2\theta}{2\cos\theta}\cdot 2\cos\theta\, d\theta = \int 4\sin^2\theta\, d\theta = 4\left(\frac{\theta}{2}-\frac{\sin(2\theta)}{4}\right) = 2\theta - \sin(2\theta) + C$$

Convert: $\theta = \arcsin\left(\frac{x}{2}\right)$, $\sin(2\theta) = 2\sin\theta\cos\theta = 2\cdot\frac{x}{2}\cdot\frac{\sqrt{4-x^2}}{2} = \frac{x\sqrt{4-x^2}}{2}$.

$$= 2\arcsin\left(\frac{x}{2}\right) - \frac{x\sqrt{4-x^2}}{2} + C$$

8. $\displaystyle\int \frac{dx}{\sqrt{x^2+4}}$, $x = 2\tan\theta$

$dx = 2\sec^2\theta\, d\theta$, $\sqrt{x^2+4} = 2\sec\theta$.

$$\int \frac{2\sec^2\theta\, d\theta}{2\sec\theta} = \int \sec\theta\, d\theta = \ln|\sec\theta + \tan\theta| + C$$

Convert: $\tan\theta = \frac{x}{2}$, $\sec\theta = \frac{\sqrt{x^2+4}}{2}$.

$$= \ln\left|\frac{\sqrt{x^2+4}+x}{2}\right| + C = \ln\left(\sqrt{x^2+4}+x\right) + C$$

(absorbing the constant $-\ln 2$ into $C$).

9. $\displaystyle\int \frac{dx}{\sqrt{x^2+9}}$, $x = 3\tan\theta$

$dx = 3\sec^2\theta\, d\theta$, $\sqrt{x^2+9} = 3\sec\theta$.

$$\int \sec\theta\, d\theta = \ln|\sec\theta + \tan\theta| + C$$

Convert: $\tan\theta = \frac{x}{3}$, $\sec\theta = \frac{\sqrt{x^2+9}}{3}$.

$$= \ln\left(\sqrt{x^2+9}+x\right) + C$$

10. $\displaystyle\int \frac{dx}{(x^2+4)^{3/2}}$, $x = 2\tan\theta$

$dx = 2\sec^2\theta\, d\theta$, $(x^2+4)^{3/2} = (4\sec^2\theta)^{3/2} = 8\sec^3\theta$.

$$\int \frac{2\sec^2\theta\, d\theta}{8\sec^3\theta} = \frac{1}{4}\int \cos\theta\, d\theta = \frac{1}{4}\sin\theta + C$$

Convert: from the triangle (opposite $= x$, adjacent $= 2$, hypotenuse $= \sqrt{x^2+4}$), $\sin\theta = \dfrac{x}{\sqrt{x^2+4}}$.

$$= \frac{x}{4\sqrt{x^2+4}} + C$$

11. $\displaystyle\int \frac{dx}{\sqrt{x^2-4}}$, $x = 2\sec\theta$

$dx = 2\sec\theta\tan\theta\, d\theta$, $\sqrt{x^2-4} = 2\tan\theta$.

$$\int \frac{2\sec\theta\tan\theta\, d\theta}{2\tan\theta} = \int \sec\theta\, d\theta = \ln|\sec\theta+\tan\theta| + C$$

Convert: $\sec\theta = \frac{x}{2}$, $\tan\theta = \frac{\sqrt{x^2-4}}{2}$.

$$= \ln\left(x+\sqrt{x^2-4}\right) + C$$

12. $\displaystyle\int \frac{dx}{x^2\sqrt{x^2-9}}$, $x = 3\sec\theta$

$dx = 3\sec\theta\tan\theta\, d\theta$, $\sqrt{x^2-9} = 3\tan\theta$, $x^2 = 9\sec^2\theta$.

$$\int \frac{3\sec\theta\tan\theta\, d\theta}{9\sec^2\theta\cdot 3\tan\theta} = \int \frac{d\theta}{9\sec\theta} = \frac{1}{9}\int \cos\theta\, d\theta = \frac{1}{9}\sin\theta + C$$

Convert: from the triangle (adjacent $= 3$, hypotenuse $= x$, opposite $= \sqrt{x^2-9}$), $\sin\theta = \dfrac{\sqrt{x^2-9}}{x}$.

$$= \frac{\sqrt{x^2-9}}{9x} + C$$

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Challenge

13. $\displaystyle\int \sqrt{9-x^2}\, dx$, $x = 3\sin\theta$

$dx = 3\cos\theta\, d\theta$, $\sqrt{9-x^2} = 3\cos\theta$.

$$\int 3\cos\theta\cdot 3\cos\theta\, d\theta = 9\int \cos^2\theta\, d\theta = 9\left(\frac{\theta}{2}+\frac{\sin\theta\cos\theta}{2}\right) + C$$

Convert: $\theta = \arcsin\left(\frac{x}{3}\right)$, $\sin\theta = \frac{x}{3}$, $\cos\theta = \frac{\sqrt{9-x^2}}{3}$.

$$= \frac{9}{2}\arcsin\left(\frac{x}{3}\right) + \frac{9}{2}\cdot\frac{x}{3}\cdot\frac{\sqrt{9-x^2}}{3} + C = \frac{9}{2}\arcsin\left(\frac{x}{3}\right) + \frac{x\sqrt{9-x^2}}{2} + C$$

14. $\displaystyle\int x^2\sqrt{9-x^2}\, dx$, $x = 3\sin\theta$

$x^2 = 9\sin^2\theta$, $\sqrt{9-x^2} = 3\cos\theta$, $dx = 3\cos\theta\, d\theta$.

$$\int 9\sin^2\theta\cdot 3\cos\theta\cdot 3\cos\theta\, d\theta = 81\int \sin^2\theta\cos^2\theta\, d\theta$$

Using the Unit 38 technique, $\sin^2\theta\cos^2\theta = \dfrac{1}{4}\sin^2(2\theta)$:

$$81\cdot\frac{1}{4}\int\sin^2(2\theta)\,d\theta = \frac{81}{4}\int\frac{1-\cos(4\theta)}{2}\,d\theta = \frac{81}{8}\left[\theta - \frac{\sin(4\theta)}{4}\right]+C$$

$$= \frac{81\theta}{8} - \frac{81}{32}\sin(4\theta)+C$$

Convert: $\theta = \arcsin\left(\frac{x}{3}\right)$. We need $\sin(4\theta) = 2\sin(2\theta)\cos(2\theta)$.

$\sin(2\theta) = 2\sin\theta\cos\theta = 2\cdot\frac{x}{3}\cdot\frac{\sqrt{9-x^2}}{3} = \dfrac{2x\sqrt{9-x^2}}{9}$.

$\cos(2\theta) = 1-2\sin^2\theta = 1-\dfrac{2x^2}{9} = \dfrac{9-2x^2}{9}$.

$$\sin(4\theta) = 2\cdot\frac{2x\sqrt{9-x^2}}{9}\cdot\frac{9-2x^2}{9} = \frac{4x\sqrt{9-x^2}(9-2x^2)}{81}$$

Substituting back:

$$\frac{81}{32}\sin(4\theta) = \frac{81}{32}\cdot\frac{4x\sqrt{9-x^2}(9-2x^2)}{81} = \frac{x\sqrt{9-x^2}(9-2x^2)}{8}$$

Final answer:

$$\int x^2\sqrt{9-x^2}\,dx = \frac{81}{8}\arcsin\left(\frac{x}{3}\right) - \frac{x(9-2x^2)\sqrt{9-x^2}}{8}+C$$

15. $\displaystyle\int\frac{dx}{\sqrt{x^2+16}}$, $x = 4\tan\theta$

$dx = 4\sec^2\theta\,d\theta$, $\sqrt{x^2+16} = 4\sec\theta$.

$$\int\sec\theta\,d\theta = \ln|\sec\theta+\tan\theta|+C$$

Convert: $\tan\theta = \frac{x}{4}$, $\sec\theta = \frac{\sqrt{x^2+16}}{4}$.

$$= \ln\left(\sqrt{x^2+16}+x\right)+C$$

16. $\displaystyle\int\frac{dx}{(x^2+9)^{3/2}}$, $x = 3\tan\theta$

$dx = 3\sec^2\theta\,d\theta$, $(x^2+9)^{3/2} = (9\sec^2\theta)^{3/2} = 27\sec^3\theta$.

$$\int\frac{3\sec^2\theta\,d\theta}{27\sec^3\theta} = \frac{1}{9}\int\cos\theta\,d\theta = \frac{1}{9}\sin\theta+C$$

Convert: $\sin\theta = \dfrac{x}{\sqrt{x^2+9}}$.

$$= \frac{x}{9\sqrt{x^2+9}} + C$$

17. $\displaystyle\int \frac{dx}{x^2\sqrt{x^2-4}}$, $x = 2\sec\theta$

$dx = 2\sec\theta\tan\theta\,d\theta$, $\sqrt{x^2-4} = 2\tan\theta$, $x^2 = 4\sec^2\theta$.

$$\int \frac{2\sec\theta\tan\theta\,d\theta}{4\sec^2\theta\cdot 2\tan\theta} = \int \frac{d\theta}{4\sec\theta} = \frac{1}{4}\int \cos\theta\,d\theta = \frac{1}{4}\sin\theta + C$$

Convert: from the triangle (adjacent $= 2$, hypotenuse $= x$, opposite $= \sqrt{x^2-4}$), $\sin\theta = \dfrac{\sqrt{x^2-4}}{x}$.

$$= \frac{\sqrt{x^2-4}}{4x} + C$$

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Applied

18. $\displaystyle\int_{-2}^{2} \sqrt{4-x^2}\,dx$

Following the same steps as Problem 13 (scaled to $a = 2$ instead of $a = 3$), the antiderivative is:

$$2\arcsin\left(\frac{x}{2}\right) + \frac{x\sqrt{4-x^2}}{2}$$

Evaluate at $x = 2$: $2\arcsin(1) + \dfrac{2\cdot 0}{2} = 2\cdot\dfrac{\pi}{2} + 0 = \pi$.

Evaluate at $x = -2$: $2\arcsin(-1) + \dfrac{-2\cdot 0}{2} = -\pi + 0 = -\pi$.

$$\pi - (-\pi) = 2\pi$$

Answer: $2\pi$. This matches the known area of a semicircle of radius 2: $\dfrac{1}{2}\pi r^2 = \dfrac{1}{2}\pi(4) = 2\pi$ — confirmed!

19. $\displaystyle\int_{0}^{3} x^2\sqrt{9-x^2}\,dx$

Using the antiderivative from Problem 14: $\dfrac{81}{8}\arcsin\left(\dfrac{x}{3}\right) - \dfrac{x(9-2x^2)\sqrt{9-x^2}}{8}$.

At $x = 3$: $\arcsin(1) = \dfrac{\pi}{2}$; the second term has a factor of $\sqrt{9-9} = 0$, so it vanishes. Value $= \dfrac{81}{8}\cdot\dfrac{\pi}{2} = \dfrac{81\pi}{16}$.

At $x=0$: both terms are 0.

$$\frac{81\pi}{16}-0=\frac{81\pi}{16}$$

Answer: $\dfrac{81\pi}{16}$.

20. $\displaystyle\int_0^4 \frac{dx}{\sqrt{x^2+16}}$

Using the antiderivative from Problem 15: $\ln\left(\sqrt{x^2+16}+x\right)$.

At $x=4$: $\ln\left(\sqrt{32}+4\right)=\ln\left(4\sqrt{2}+4\right)=\ln\left(4(\sqrt{2}+1)\right)=\ln 4+\ln(\sqrt{2}+1)$.

At $x=0$: $\ln\left(\sqrt{16}+0\right)=\ln 4$.

$$\left[\ln 4+\ln(\sqrt{2}+1)\right]-\ln 4=\ln(\sqrt{2}+1)$$

Answer: $\ln(\sqrt{2}+1)\approx 0.881$.

21. $\displaystyle\int_3^{3\sqrt{2}} \frac{dx}{x^2\sqrt{x^2-9}}$

Following the same steps as Problem 12 (with $a=3$), the antiderivative is $\dfrac{\sqrt{x^2-9}}{9x}$.

At $x=3\sqrt{2}$: $\sqrt{(3\sqrt{2})^2-9}=\sqrt{18-9}=\sqrt{9}=3$. Value $=\dfrac{3}{9(3\sqrt{2})}=\dfrac{3}{27\sqrt{2}}=\dfrac{1}{9\sqrt{2}}=\dfrac{\sqrt{2}}{18}$.

At $x=3$: $\sqrt{9-9}=0$. Value $=0$.

$$\frac{\sqrt{2}}{18}-0=\frac{\sqrt{2}}{18}$$

Answer: $\dfrac{\sqrt{2}}{18}$.